JSC370 Final Project: Country-Level Cancer Mortality

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GitHub repository: https://github.com/Terrachizoid/JSC370Midterm

1 Introduction

Cancer mortality varies substantially across countries and is shaped by a complex combination of behavioral, environmental, demographic, economic, and healthcare-related factors. Understanding whether broad country-level indicators are associated with these differences can help identify which factors are most informative at the population level, while also highlighting the limitations of using aggregated international health data for prediction.

This project examines cross-country variation in cancer mortality using publicly available health and development indicators. The main data source is the **World Health Organization (WHO) Global Health Observatory API**, which provides internationally comparable measures of cancer mortality and several major risk factors. To supplement these data, the **World Bank API** is used to obtain GDP per capita and life expectancy at birth. Together, these sources allow the construction of a country-level dataset combining mortality, lifestyle risk factors, environmental exposure, socioeconomic conditions, and healthcare resources.

The original goal was to build a recent country-year panel and study changes over time. However, after retrieving the indicators, it became clear that year coverage was highly uneven across variables and countries. Some indicators were reported regularly, while others appeared only sporadically within the target period. To obtain a dataset with broader country coverage and a common set of predictors, the analysis was therefore reformulated as a **cross-sectional study**, using the most recent available observation for each country between 2014 and 2023.

The main research question is: **To what extent can country-level lifestyle risk factors, socioeconomic conditions, environmental exposure, demographic structure, and healthcare resources explain variation in cancer mortality across countries?**

I hypothesized that smoking, obesity, alcohol consumption, and physical inactivity would be positively associated with cancer mortality. I also expected life expectancy to be related to mortality because it partly reflects population age structure, longevity, and overall health-system development. At the same time, I expected predictive performance to be limited, since cancer mortality is influenced by many overlapping processes that are not fully captured by a small set of aggregate country-level indicators.

2 Data and Preprocessing

2.1 Data sources and study design

Data were collected from two public APIs. The **WHO Global Health Observatory API** was used to retrieve the original outcome variable, age-standardized all-cancer mortality, five cancer-type-specific mortality outcomes, and several candidate predictors: smoking prevalence, obesity prevalence, alcohol consumption, prevalence of insufficient physical activity, PM2.5 exposure, medical doctors per 10,000 population, cervical cancer screening among women aged 30–49, and chronic hepatitis B prevalence. The **World Bank API** was used to obtain GDP per capita, life expectancy at birth, and urban population share. These variables were selected to represent behavioral risk factors, environmental exposure, socioeconomic development, demographic structure, healthcare capacity, screening behavior, infection-related risk, and urbanization context.

Because the temporal coverage of the indicators was inconsistent across countries and variables, the analysis was designed as a **cross-sectional country-level study** rather than a panel analysis. For the main all-cancer outcome and predictor variables, observations were restricted to the 2014–2023 period and the most recent available value for each country was retained. For the five cancer-type-specific mortality outcomes, the WHO API only provided older observations for many countries, so the most recent available value was retained without applying the 2014–2023 filter. These cancer-type models are therefore treated as exploratory cross-country association models rather than causal or forward-looking prediction models.

2.2 Data acquisition and preprocessing

The code below imports the required packages, defines an indicator registry, and defines helper functions used to download, filter, and standardize the indicators from the WHO and World Bank APIs.

Table 1 summarizes the indicators included in the final analytic dataset.

	variable	source	indicator_code
0	cancer_mortality	WHO	SA_0000001807
1	smoking	WHO	M_Est_tob_curr
2	obesity	WHO	NCD_BMI_30A
3	alcohol	WHO	SA_0000001822
4	inactivity	WHO	NCD_PAA
5	pollution_pm25	WHO	SDGPM25
6	medical_doctors_per_10000	WHO	HWF_0001
7	cervical_screening_rate	WHO	NCD_CXCA_SCREENED_WITHIN_TIMEPE
8	hepatitis_b_prevalence	WHO	HEPATITIS_HBV_PREVALENCE_PER100

	variable	source	indicator_code
9	breast_cancer_mortality	WHO	SA_0000001438
10	mouth_oropharynx_cancer_mortality	WHO	SA_0000001448
11	oesophagus_cancer_mortality	WHO	SA_0000001449
12	colorectal_cancer_mortality	WHO	SA_0000001439
13	liver_cancer_mortality	WHO	SA_0000001445
14	gdp_per_capita	World Bank	NY.GDP.PCAP.CD
15	life_expectancy	World Bank	SP.DYN.LE00.IN
16	urban_population_pct	World Bank	SP.URB.TOTL.IN.ZS

2.3 Building the country-level dataset

The selected all-cancer outcome and predictor indicators were downloaded and restricted to the 2014–2023 period. Because year coverage differs across indicators, the most recent available observation for each country in the selected window was retained. The five cancer-type-specific mortality outcomes were downloaded separately and kept at their most recent available year because restricting them to 2014–2023 removes nearly all observations.

This produces a cross-sectional dataset rather than a sparse country-year panel.

	variable	countries_available	label
0	cancer_mortality	180	Cancer mortality
1	smoking	167	Smoking prevalence (%)
2	obesity	199	Obesity prevalence (%)
3	alcohol	194	Alcohol consumption
4	inactivity	195	Physical inactivity prevalence (%)
5	pollution_pm25	195	PM2.5 exposure
6	medical_doctors_per_10000	193	Medical doctors (per 10,000)
7	cervical_screening_rate	195	Cervical screening rate (%)
8	hepatitis_b_prevalence	167	Chronic hepatitis B prevalence (%)
9	breast_cancer_mortality	192	Breast cancer mortality
10	mouth_oropharynx_cancer_mortality	192	Mouth/oropharynx cancer mortality
11	oesophagus_cancer_mortality	191	Oesophagus cancer mortality
12	colorectal_cancer_mortality	192	Colon and rectum cancer mortality
13	liver_cancer_mortality	192	Liver cancer mortality
14	gdp_per_capita	261	GDP per capita (current US$)
15	life_expectancy	261	Life expectancy at birth (years)
16	urban_population_pct	261	Urban population (%)

The table below records the year range of the five cancer-type-specific mortality outcomes. This is included because these outcomes are not temporally aligned with the 2014–2023 predictor

window and should be interpreted as exploratory country-level comparisons rather than causal prediction targets.

	variable	label	min_year	median_year
0	breast_cancer_mortality	Breast cancer mortality	2002	2004.0
1	mouth_oropharynx_cancer_mortality	Mouth/oropharynx cancer mortality	2002	2004.0
2	oesophagus_cancer_mortality	Oesophagus cancer mortality	2002	2004.0
3	colorectal_cancer_mortality	Colon and rectum cancer mortality	2002	2004.0
4	liver_cancer_mortality	Liver cancer mortality	2002	2004.0

Overall cancer mortality was used as the base table because it is the original outcome of interest. Cancer-type-specific mortality outcomes and predictor tables were then merged by country.

	variable	missing_before_imputation
0	country	0
1	cancer_mortality	0
2	breast_cancer_mortality	0
3	mouth_oropharynx_cancer_mortality	0
4	oesophagus_cancer_mortality	6
5	colorectal_cancer_mortality	0
6	liver_cancer_mortality	5
7	smoking	22
8	alcohol	0
9	obesity	0
10	inactivity	0
11	gdp_per_capita	5
12	life_expectancy	0
13	pollution_pm25	0
14	medical_doctors_per_10000	0
15	cervical_screening_rate	0
16	hepatitis_b_prevalence	27
17	urban_population_pct	0

The table below checks raw availability for the newly added predictor variables and cancer-type-specific mortality outcomes before median imputation. This helps separate real coverage from values later filled during preprocessing.

	count	min	median	max
cervical_screening_rate	180.0	0.00	19.50	100.0

	count	min	median	max
hepatitis_b_prevalence	153.0	0.10	2.40	13.9
urban_population_pct	180.0	15.06	61.64	100.0
breast_cancer_mortality	180.0	0.00	10.10	39.1
mouth_oropharynx_cancer_mortality	180.0	0.00	3.50	41.6
oesophagus_cancer_mortality	174.0	0.00	2.40	50.7
colorectal_cancer_mortality	180.0	0.00	8.85	41.8
liver_cancer_mortality	175.0	0.30	5.80	117.6

These added variables support a later cancer-type-specific modeling extension. The clean WHO mortality outcomes available from the current API search are breast cancer, mouth/oropharynx cancer, oesophagus cancer, colon and rectum cancer, and liver cancer. UV radiation and latitude were removed from the analytic dataset because the corresponding skin cancer mortality outcome was not available in this WHO indicator set.

2.4 Missing data and analytic variables

Remaining numeric missing values were imputed using variable medians. This is a simple and transparent strategy that preserves country coverage for both exploratory analysis and predictive modeling while avoiding a substantial loss of observations from complete-case analysis.

	variable	missing_after_imputation
0	country	0
1	cancer_mortality	0
2	breast_cancer_mortality	0
3	mouth_oropharynx_cancer_mortality	0
4	oesophagus_cancer_mortality	0
5	colorectal_cancer_mortality	0
6	liver_cancer_mortality	0
7	smoking	0
8	alcohol	0
9	obesity	0
10	inactivity	0
11	gdp_per_capita	0
12	life_expectancy	0
13	pollution_pm25	0
14	medical_doctors_per_10000	0
15	cervical_screening_rate	0
16	hepatitis_b_prevalence	0
17	urban_population_pct	0

The final analytic dataset contains one row per country. Table 2 reports summary statistics for the numeric variables.

	variable	mean	median	std	min	max
0	Cancer mortality	122.89	118.09	37.78	48.35	262.45
1	Breast cancer mortality	9.72	10.10	9.05	0.00	39.10
2	Mouth/oropharynx cancer mortality	5.19	3.50	6.56	0.00	41.60
3	Oesophagus cancer mortality	5.16	2.40	8.05	0.00	50.70
4	Colon and rectum cancer mortality	10.08	8.85	6.37	0.00	41.80
5	Liver cancer mortality	11.01	5.80	15.26	0.30	117.60
6	Smoking prevalence (%)	19.37	17.70	13.53	0.00	74.70
7	Alcohol consumption	6.01	4.15	5.41	0.00	24.60
8	Obesity prevalence (%)	21.80	21.34	11.89	1.06	71.65
9	Physical inactivity prevalence (%)	27.90	25.08	13.74	2.70	73.60
10	GDP per capita (current US$)	16776.50	6299.99	23307.51	250.63	133230.62
11	Life expectancy at birth (years)	73.11	73.60	6.98	54.46	84.16
12	PM2.5 exposure	22.66	17.67	14.81	5.20	84.04
13	Medical doctors (per 10,000)	20.26	16.76	17.30	0.38	85.05
14	Cervical screening rate (%)	35.43	19.50	32.72	0.00	100.00
15	Chronic hepatitis B prevalence (%)	3.56	2.40	3.11	0.10	13.90
16	Urban population (%)	59.85	61.64	21.93	15.06	100.00

3 Exploratory Data Analysis

Exploratory data analysis included histograms, scatterplots, correlation summaries, and boxplots to assess variable distributions, inspect bivariate relationships with cancer mortality, and identify skewness or outliers that might affect modeling. Interactive visualizations were also created as a companion website component to support geographic interpretation, country comparison, and exploration of selected predictors.

3.1 Distribution of variables

Figure 1 shows the distributions of the numeric variables in the cleaned dataset.

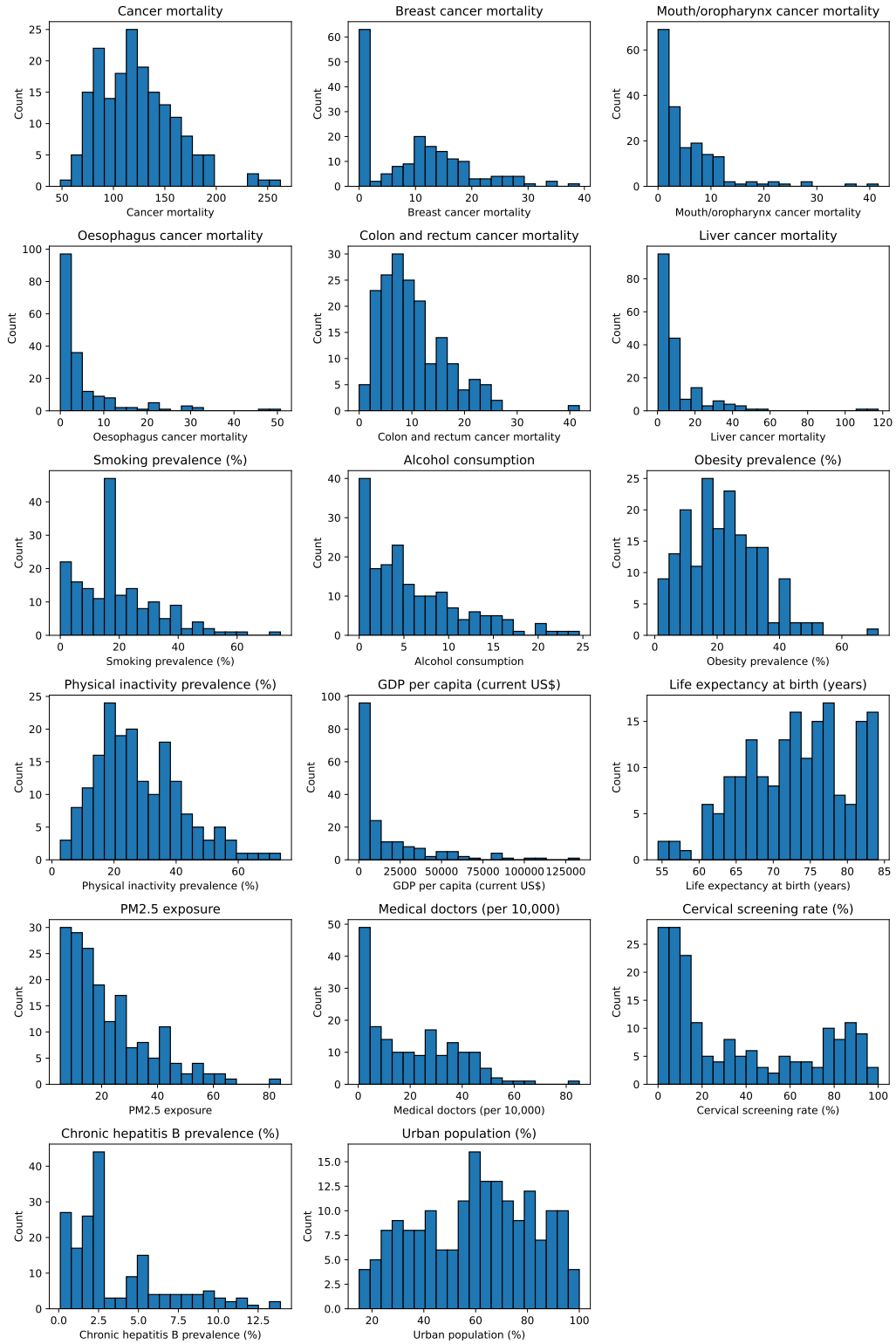


Figure 1: Figure 1. Distributions of cancer mortality and predictor variables across countries. Several variables show skewness and noticeable cross-country heterogeneity, motivating transformations and careful modeling choices.

Most variables show some degree of right-skewness. GDP per capita is the clearest example, reflecting the large gap between lower-income and higher-income countries, while life expectancy appears more concentrated with less extreme spread. This suggests that transformations such as a log scale may be useful for selected predictors in the predictive modeling stage.

3.2 Bivariate relationships with cancer mortality

Figure 2 shows scatterplots of cancer mortality against the main predictors.

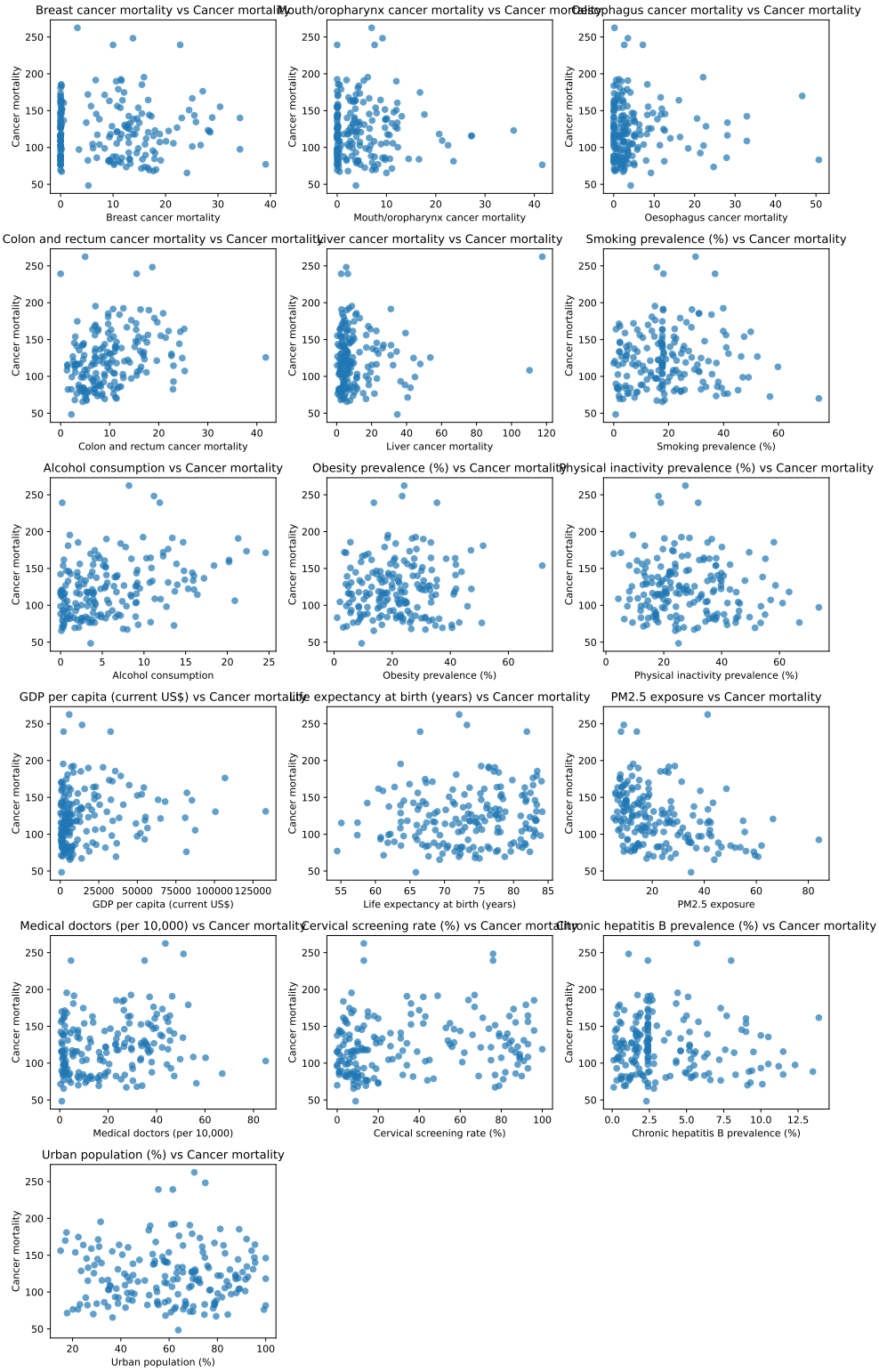


Figure 2: Figure 2. Scatterplots of cancer mortality against selected predictors. Most relationships appear weak to moderate and are accompanied by substantial variability, suggesting that no single variable explains country-level cancer mortality particularly well on its own.

The scatterplots suggest weak to moderate relationships overall. Smoking, alcohol consumption, and obesity show slight upward patterns, while life expectancy appears to capture broader demographic and development differences that may also matter for cancer mortality. Even so, each pattern is accompanied by considerable noise. This is not surprising in a country-level analysis, where cancer mortality is shaped by many overlapping demographic, behavioral, environmental, and healthcare factors.

3.3 Correlation structure

Figure 3 presents the correlation matrix for the numeric variables.

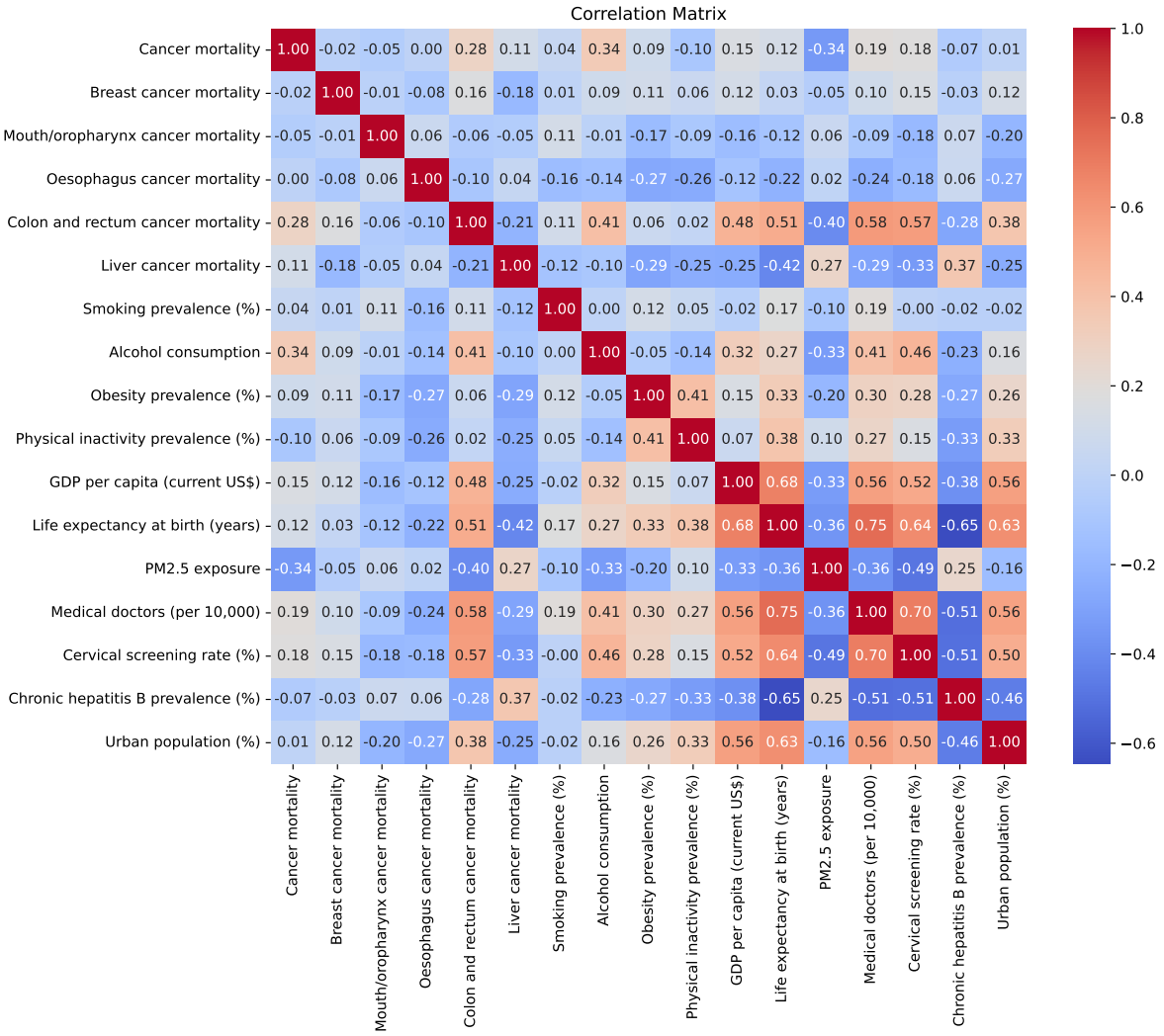


Figure 3: Figure 3. Correlation matrix of the numeric variables. Most correlations with cancer mortality are modest, while some stronger associations appear among predictors themselves.

The correlation matrix reinforces the same general pattern. Correlations between cancer mortality and individual predictors are mostly modest, although smoking, alcohol consumption, and obesity tend to have small positive associations. Life expectancy also appears as a broader population-health measure rather than a strong standalone driver. GDP per capita is positively related to physician density, which is consistent with the idea that wealthier countries tend to have greater healthcare capacity.

One noteworthy result is the negative correlation between PM2.5 exposure and cancer mortality. This should not be interpreted as evidence of a protective effect. A more plausible explanation

is confounding from factors such as life expectancy, age structure, and differences in reporting quality across countries.

3.4 Outlier inspection

Figure 4 shows boxplots for the numeric variables.

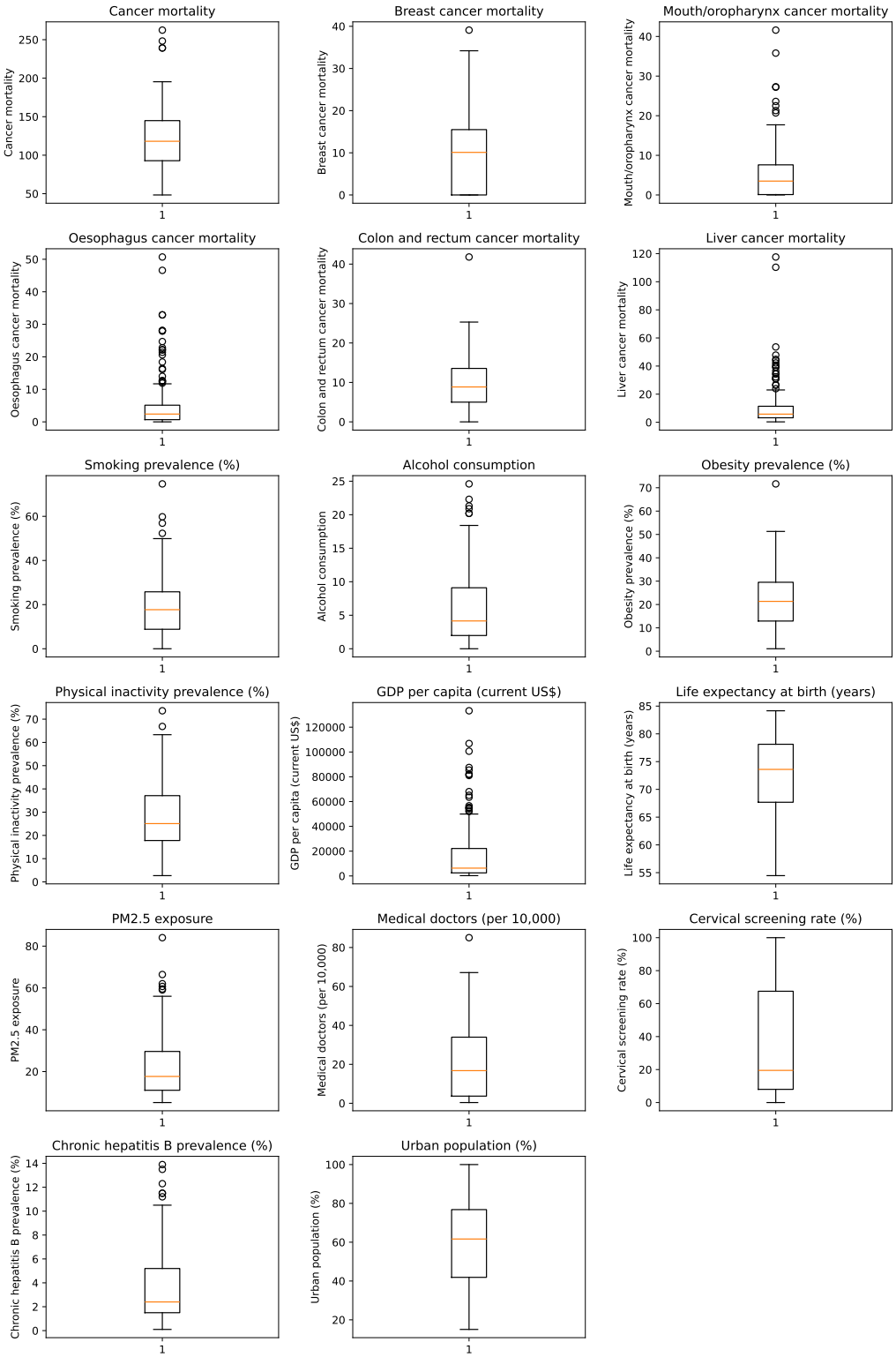


Figure 4: Figure 4. Boxplots for cancer mortality and predictor variables. Several variables contain high-end outliers, which may motivate transformations or robust modeling choices.

Several predictors contain noticeable high-end outliers, especially GDP per capita and physician density. This reflects the wide gap between countries rather than obvious data entry errors, but it suggests that careful treatment of skewed variables will matter in later modeling.

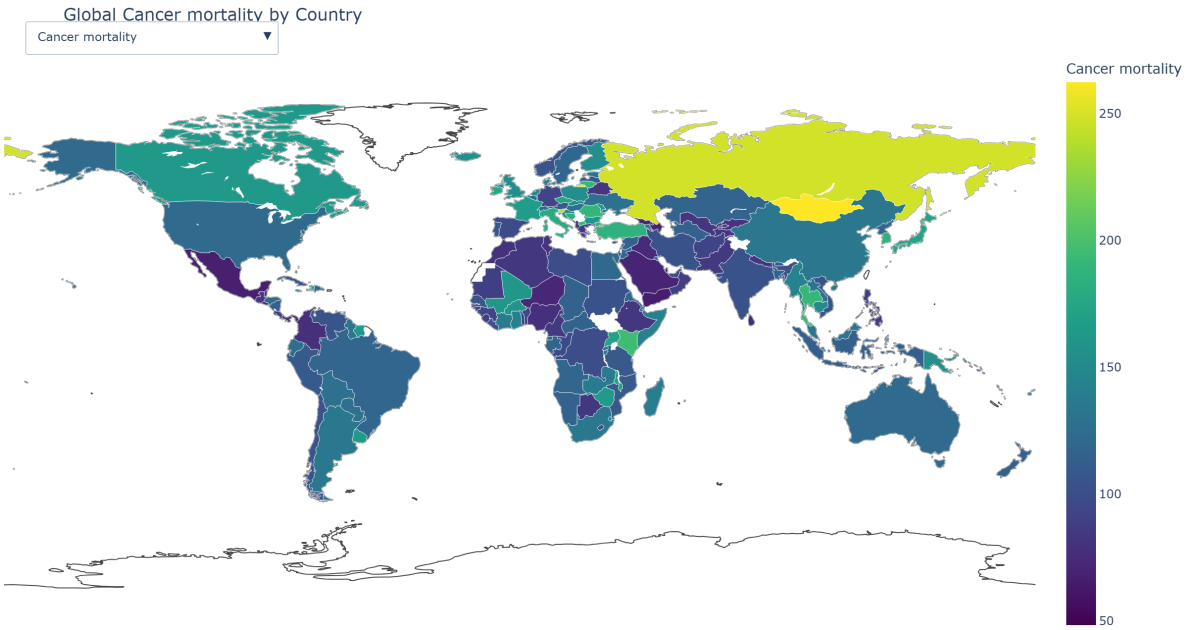
3.5 Interactive visualizations

The following interactive figures extend the exploratory analysis using the cleaned country-level dataset developed for the project. They are presented in report style, with each visualization paired with a short caption and interpretation. The goal is not to introduce new modeling results, but to give readers multiple interactive ways to examine the geographic distribution of the data, the relationship between selected predictors and cancer mortality, and the countries that rank highest or lowest on each measure.

The PDF version shows static versions of the interactive website figures so that the report remains readable in print. The full interactive versions are available on the website.

3.5.1 Interactive Figure 1. Global choropleth map of selected indicators

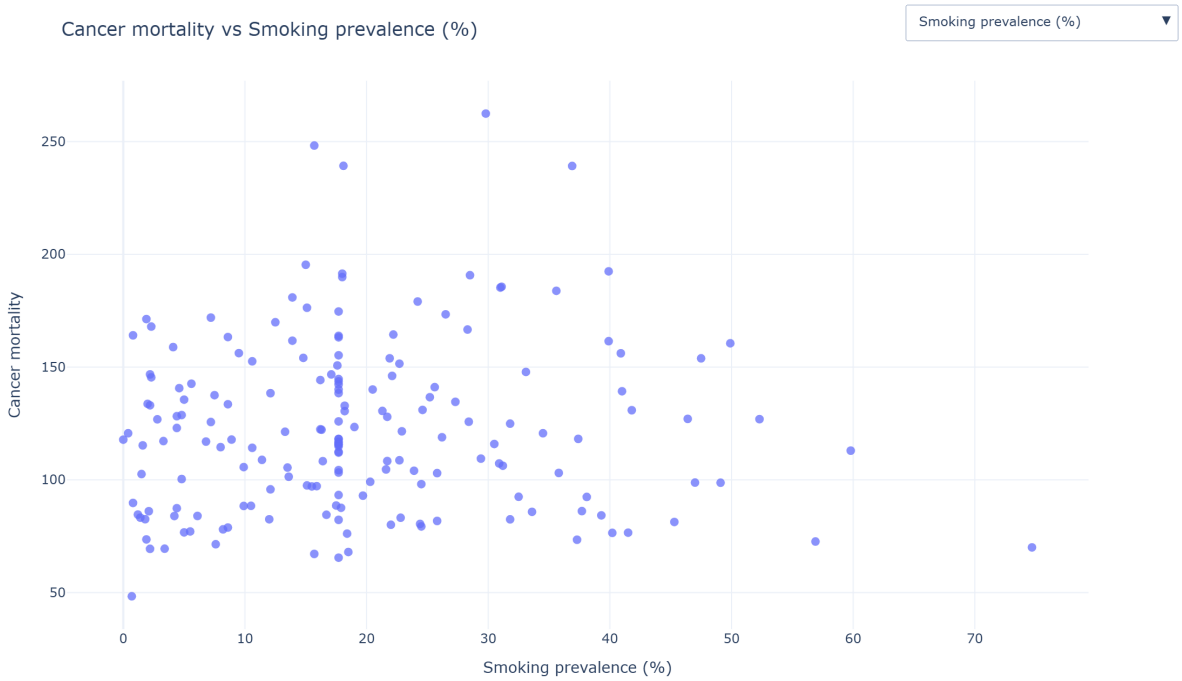
This figure provides a geographic view of the dataset. The dropdown menu allows the user to switch between all-cancer mortality, cancer-type-specific mortality outcomes, and several predictors, including smoking prevalence, alcohol consumption, obesity prevalence, physical inactivity, GDP per capita, life expectancy, cervical screening, hepatitis B prevalence, and urban population share. Hover text displays the country name, ISO-3 code, and the selected value. This plot is useful for identifying broad regional structure and for comparing both overall mortality and cancer-type-specific mortality patterns across countries.



Caption. *Interactive choropleth map of cancer mortality and selected country-level predictors.* Use the dropdown menu to switch the displayed variable. Darker shading indicates larger values for the selected measure, and hovering over a country reveals its name, ISO-3 code, and numeric value.

3.5.2 Interactive Figure 2. Cancer mortality versus selected predictors

This scatter plot fixes cancer mortality on the y-axis and lets the reader choose a single predictor for the x-axis. That design is intentionally simpler than a two-dropdown version: it avoids synchronization problems and keeps the interpretation focused on the main project question, namely how each candidate predictor relates to cancer mortality. Each point is a country, and hovering reveals the country name, ISO-3 code, and both variable values.



Caption. *Interactive scatter plot of cancer mortality against selected predictors.* The dropdown menu changes the predictor shown on the x-axis while cancer mortality remains fixed on the y-axis. This view makes it easier to compare whether smoking, alcohol consumption, obesity, physical inactivity, GDP per capita, life expectancy, cervical screening, hepatitis B prevalence, or urban population share show stronger or weaker country-level associations with mortality.

3.5.3 Interactive Figure 3. Country rankings for selected indicators

The third figure is an interactive ranking table. It allows the user to switch between variables and inspect the countries with either the highest or lowest observed values. This complements the map and scatter plot by giving an exact lookup of country names and values rather than only a visual impression.

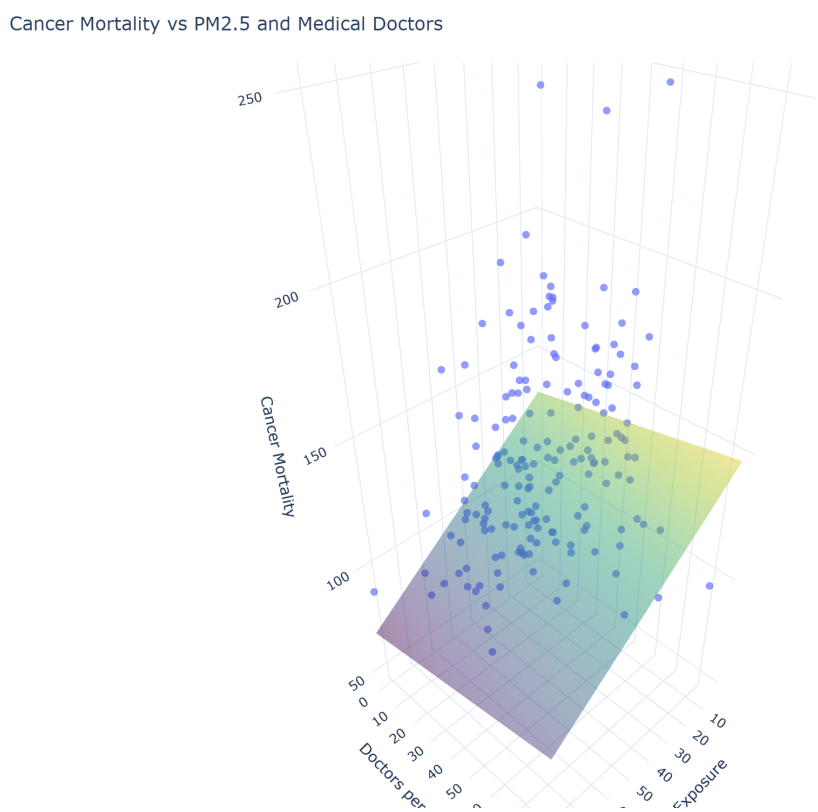
	Rank	Country	ISO-3	Cancer mortality
0	1	Mongolia	MNG	262.448127
1	2	Russian Federation	RUS	248.271735
2	3	Slovenia	SVN	239.270063
3	4	Kiribati	KIR	239.221815
4	5	Kenya	KEN	195.388000
5	6	North Macedonia	MKD	192.473933
6	7	Thailand	THA	191.464763

	Rank	Country	ISO-3	Cancer mortality
7	8	Lithuania	LTU	190.767123
8	9	Romania	ROU	189.912971
9	10	Korea, Republic of	KOR	185.646510

Caption. *Interactive ranking table for country-level indicators.* The dropdown menu switches between variables and ordering rules, allowing the reader to inspect the countries with the ten highest or ten lowest values for the selected measure.

3.5.4 Exploratory figure. 3D scatter plot with fitted plane

This final figure is not part of the three required HW5 visuals, but it is included as an exploratory extension. It plots cancer mortality against PM2.5 exposure and medical doctor density, then overlays a fitted linear plane. The purpose is not to claim a causal mechanism or to present a final predictive model, but to illustrate that some of the most surprising bivariate findings become easier to discuss when viewed jointly.



Caption. *Exploratory 3D scatter plot of cancer mortality, PM2.5 exposure, and medical doctor density with fitted linear plane.* The fitted plane indicates a positive association between doctor density and recorded cancer mortality, alongside a negative association between PM2.5 exposure and mortality. These patterns are counterintuitive and should not be interpreted causally; they more plausibly reflect confounding by development, healthcare access, life expectancy, and reporting quality.

4 Predictive Modeling

The modeling section has two parts. **Part 1** keeps the original all-cancer mortality task, where the goal is to evaluate whether the broad country-level indicators explain variation in overall cancer mortality. **Part 2** extends the analysis to five cancer-type-specific outcomes. The second part is intentionally framed as exploratory association modeling because the cancer-type-specific WHO mortality outcomes are older than the 2014–2023 predictor window.

4.1 Part 1: Overall cancer mortality model

4.1.1 Modeling strategy

To extend the exploratory analysis into the predictive stage of the project, I first compared three regression models for overall cancer mortality: a baseline **ordinary least squares linear regression**, a **random forest regressor**, and an **XGBoost regressor**. The same outcome and predictor set were used across models so that performance differences would reflect model flexibility rather than differences in inputs.

The outcome variable was **overall cancer mortality**. The predictor set included smoking, obesity, alcohol consumption, physical inactivity, GDP per capita, PM2.5 exposure, medical doctor density, and life expectancy. For the linear model, GDP per capita, PM2.5 exposure, and doctor density were log-transformed to reduce skewness. The tree-based models were fit on the original scale because they do not require linearity or normally distributed predictors.

Model performance was assessed with **5-fold cross-validation** using three metrics: **(R^2)**, **RMSE**, and **MAE**. To provide a useful benchmark, a null model that predicts the mean cancer mortality was also included in the final comparison table.

4.1.2 Baseline models

The linear regression model serves as an interpretable baseline, while the initial random forest model provides a constrained nonlinear benchmark with shallow trees and minimum node-size restrictions to reduce overfitting.

	Model	CV R^2	CV R^2 SD	CV RMSE	CV MAE	Train R^2
0	Null model	-0.0537	0.0431	37.8647	29.8722	NaN
1	Linear regression	0.0534	0.1112	35.8185	27.6257	NaN
2	Random forest (constrained)	0.0162	0.1007	36.5395	28.0472	0.4792

These baseline results provide a reference point for judging whether more flexible models meaningfully improve prediction or mainly increase variance.

4.1.3 Random forest tuning

Random forest hyperparameters were tuned in two stages. A small **randomized search** was first used to explore a broad but manageable parameter space. A more focused **grid search** was then conducted around the better-performing region suggested by the randomized search. This two-stage strategy is more efficient than a large exhaustive grid while still allowing the main bias–variance trade-offs to be explored.

	mean_test_score	std_test_score	mean_train_score	param_n_estimators	param_max_depth	p
11	0.0596	0.0866	0.3387	200	3	1
19	0.0528	0.0975	0.3917	200	5	5
15	0.0524	0.0973	0.3723	300	4	5
4	0.0501	0.0985	0.3935	300	6	1
1	0.0472	0.0810	0.4273	100	3	1
0	0.0465	0.1102	0.4108	200	6	5
14	0.0447	0.0759	0.4066	100	3	1
17	0.0441	0.0818	0.4428	100	3	5
3	0.0431	0.0786	0.2989	100	3	1
8	0.0429	0.0921	0.4369	100	4	1

	mean_test_score	std_test_score	mean_train_score	param_n_estimators	param_max_depth	p
16	0.0596	0.0866	0.3387	200	3	1
13	0.0596	0.0866	0.3387	200	3	5
48	0.0581	0.0921	0.3755	100	4	5
51	0.0581	0.0921	0.3755	100	4	1
14	0.0578	0.0863	0.3360	300	3	5
17	0.0578	0.0863	0.3360	300	3	1
12	0.0553	0.0858	0.3389	100	3	5
15	0.0553	0.0858	0.3389	100	3	1
52	0.0551	0.0962	0.3755	200	4	1

	mean_test_score	std_test_score	mean_train_score	param_n_estimators	param_max_depth	p
49	0.0551	0.0962	0.3755	200	4	5

	max_depth	max_features	min_samples_leaf	min_samples_split	n_estimators
0	3	0.5	8	5	200

The random forest tuning results indicate modest restrictions on tree depth, node size, and feature subsampling improve generalization.

Increasing the number of trees beyond 100 did not improve predictive performance. While random forests theoretically reduce variance as the number of trees increases, this effect diminishes once a sufficient number of trees has been reached. In this dataset, performance differences between 100, 200, and 300 trees were negligible relative to cross-validation variability, indicating that model performance had already plateaued.

Instead, hyperparameters controlling tree complexity and feature subsampling (e.g., maximum depth, minimum samples per leaf, and max_features) had a larger impact on generalization performance, likely by reducing correlation between trees and preventing overfitting.

4.1.4 XGBoost modeling and tuning

XGBoost was included as a second tree-based model with a different inductive bias. Unlike random forests, which average many independently grown trees, XGBoost builds trees sequentially so that each new tree attempts to correct the errors of the previous ensemble. This can improve performance when the underlying structure is complex, but it also makes the method more prone to overfitting on small or noisy datasets.

	Model	CV R^2	CV R^2 SD	CV RMSE	CV MAE	Train R^2
0	XGBoost (default)	-0.2954	0.205	41.6182	32.2278	0.9103

The default XGBoost model shows clear overfitting: its training (R^2) is high, but its cross-validated (R^2) remains much lower. This indicates that the model captured patterns specific to the training folds more strongly than patterns that generalize across countries.

A focused grid search was used for XGBoost rather than a very large search space. Given the small and noisy dataset, the search emphasized shallower trees, moderate numbers of boosting rounds, subsampling, and stronger regularization.

	mean_test_score	std_test_score	mean_train_score	param_n_estimators	param_max_depth	p
2	-0.0377	0.1168	0.4954	100	2	0
50	-0.0592	0.1187	0.5166	100	2	0
0	-0.0723	0.1174	0.5660	100	2	0
3	-0.0787	0.1323	0.5070	100	2	0
14	-0.0889	0.1526	0.6729	100	3	0
6	-0.0928	0.1368	0.5891	150	2	0
51	-0.0982	0.1363	0.5204	100	2	0
15	-0.0989	0.1408	0.6901	100	3	0
48	-0.0992	0.1094	0.5837	100	2	0
62	-0.1097	0.1364	0.6906	100	3	0

	colsample_bytree	learning_rate	max_depth	n_estimators	reg_lambda	subsample
0	0.7	0.05	2	100	5	0.7

4.1.5 Final model comparison for overall cancer mortality

The final comparison table summarizes cross-validated performance across the null baseline, linear regression, constrained and tuned random forest models, and default and tuned XGBoost models.

	Model	Train R^2	CV R^2	CV R^2 SD	CV RMSE	CV MAE
0	Null model	NaN	-0.0537	0.0431	37.8647	29.8722
1	Linear regression	NaN	0.0534	0.1112	35.8185	27.6257
2	Random forest (constrained)	0.4792	0.0162	0.1007	36.5395	28.0472
3	Random forest (tuned)	0.3387	0.0596	0.0866	35.7468	27.8065
4	XGBoost (default)	0.9103	-0.2954	0.2050	41.6182	32.2278
5	XGBoost (tuned)	0.4954	-0.0377	0.1168	37.4299	28.9398

4.1.6 Overall cancer model interpretation

The predictive performance of all models is summarized in the final comparison table. Overall, the non-null models performed better than the null baseline, but none achieved strong predictive accuracy. The cross-validated **(R^2)** values remained low across all approaches, suggesting that the predictors contain some signal but not enough for highly accurate country-level prediction.

The linear regression model provided a simple baseline and achieved positive but limited explanatory power. The tree-based models generally improved on the null baseline, but the

gains were still modest. The strongest warning sign came from the default XGBoost model: although it did not fall below the null baseline in cross-validation, its very high training (R^2) compared with its much lower cross-validated (R^2) indicates substantial overfitting and weak generalization.

A consistent pattern across models is that added flexibility only produced small improvements in predictive performance. Tree-based methods can capture nonlinear relationships and interactions, but in this dataset their extra flexibility also increased the risk of fitting noise. This is especially visible in the larger gaps between training and cross-validation performance for the more flexible configurations.

After hyperparameter tuning, both random forest and XGBoost models showed improved generalization. The tuned models used shallower trees, larger minimum node sizes, and subsampling strategies, which effectively reduced overfitting. This is evident from the reduction in training performance accompanied by modest improvements in cross-validated metrics.

Among all models, the tuned random forest achieved the best overall performance, although the improvement over simpler models was small. This suggests that while some nonlinear structure may exist, it is weak and difficult to exploit reliably given the available data.

4.2 Part 2: Cancer-type-specific exploratory models

The second modeling stage asks whether different cancer types show stronger associations with smaller, mechanism-informed predictor sets. Instead of fitting one large model with every available variable, each cancer-specific outcome is paired with three to four predictors that are biologically or contextually plausible for that cancer type. This avoids unnecessary feature expansion in a dataset with fewer than 200 countries.

The cancer-type outcomes use the most recent WHO observation available for each country, which is older than the 2014–2023 predictor window for many countries. For that reason, these models should be interpreted as **exploratory cross-country association models**, not causal models and not real forward-looking prediction systems.

I do not use stepwise AIC feature selection here. AIC is most natural for likelihood-based parametric models and is not directly comparable across random forests, spline-based models, and regularized regressions. Stepwise selection can also be unstable in small, correlated country-level datasets. Instead, the feature sets below are mechanism-informed, while Elastic Net provides a regularized linear model that can shrink weak predictors automatically.

Table 17: Mechanism-informed p
models.

	Cancer outcome	Predictors used
0	Breast cancer mortality	Cervical screening rate (%), Obesity prevalence (%), Alcohol consun

Table 17: Mechanism-informed p
models.

	Cancer outcome	Predictors used
1	Mouth/oropharynx cancer mortality	Smoking prevalence (%), Alcohol consumption, GDP per capita (cu
2	Oesophagus cancer mortality	Smoking prevalence (%), Alcohol consumption, Obesity prevalence (
3	Colon and rectum cancer mortality	Obesity prevalence (%), Physical inactivity prevalence (%), Alcohol
4	Liver cancer mortality	Chronic hepatitis B prevalence (%), Alcohol consumption, Obesity p

4.2.1 Candidate model families

For each cancer-specific outcome, four model families are compared under 5-fold cross-validation:

1. **Linear regression**, used as an untuned baseline.
2. **Elastic Net**, which tunes the penalty strength and the ridge/lasso mixture to stabilize linear estimation under correlated predictors.
3. **Additive spline ridge model**, used as a GAM-like model. It applies spline basis functions to each predictor and then fits a regularized additive regression. This captures smooth nonlinear relationships without allowing unrestricted interactions.
4. **Random forest**, tuned with a small local grid emphasizing shallow trees and larger leaf sizes. This provides a nonlinear interaction-capable benchmark while limiting variance.

Because several cancer-type mortality outcomes are strongly right-skewed, linear, Elastic Net, and additive spline models are fit through a log-target transformation using `log1p`. Performance metrics are still computed on the original mortality scale after inverse transformation. Random forests are fit on the raw outcome scale.

Table 18: Cross-validated performance of candidate model families for each cancer-type-spec
mortality outcome.

	Cancer outcome	Model	Train R^2	CV R^2	CV R^2 SD
0	Breast cancer mortality	Linear regression	-0.2447	-0.3241	0.1384
1	Breast cancer mortality	Elastic Net (tuned)	-0.2464	-0.3233	0.1393
2	Breast cancer mortality	Additive spline ridge (tuned)	-0.2175	-0.3547	0.1400
3	Breast cancer mortality	Random forest (tuned)	0.1234	-0.0914	0.1599
4	Mouth/oropharynx cancer mortality	Linear regression	-0.1061	-0.1522	0.0707
5	Mouth/oropharynx cancer mortality	Elastic Net (tuned)	-0.1068	-0.1522	0.0713
6	Mouth/oropharynx cancer mortality	Additive spline ridge (tuned)	0.0629	-0.1111	0.0579
7	Mouth/oropharynx cancer mortality	Random forest (tuned)	0.2974	0.1043	0.0780
8	Oesophagus cancer mortality	Linear regression	-0.0292	-0.0694	0.0784
9	Oesophagus cancer mortality	Elastic Net (tuned)	-0.0306	-0.0703	0.0782

Table 18: Cross-validated performance of candidate model families for each cancer-type-spec
mortality outcome.

	Cancer outcome	Model	Train R^2	CV R^2	CV R^2 SD
10	Oesophagus cancer mortality	Additive spline ridge (tuned)	-0.0134	-0.0895	0.0691
11	Oesophagus cancer mortality	Random forest (tuned)	0.2333	0.0279	0.1467
12	Colon and rectum cancer mortality	Linear regression	0.1963	0.0148	0.3571
13	Colon and rectum cancer mortality	Elastic Net (tuned)	0.2072	0.1810	0.0880
14	Colon and rectum cancer mortality	Additive spline ridge (tuned)	0.3626	0.3158	0.1202
15	Colon and rectum cancer mortality	Random forest (tuned)	0.6382	0.3801	0.1318
16	Liver cancer mortality	Linear regression	0.1173	0.1301	0.0918
17	Liver cancer mortality	Elastic Net (tuned)	0.1160	0.1294	0.0915
18	Liver cancer mortality	Additive spline ridge (tuned)	0.1409	0.1222	0.0964
19	Liver cancer mortality	Random forest (tuned)	0.3427	0.2067	0.0871

The cancer-type comparison shows a clearer pattern than the single all-cancer model. The tuned random forest is selected most often, suggesting that nonlinear relationships and interactions are useful for these country-level cancer outcomes. However, the strength of prediction differs substantially by cancer type. Colon and rectum cancer and liver cancer show stronger cross-validated performance than breast, oesophagus, and mouth/oropharynx cancer, which suggests that some cancer outcomes align more clearly with the measured country-level risk profiles than others. This supports the decision to model cancer types separately: aggregating all cancers together mixes distinct mechanisms and can blur signals that are more visible for individual cancer outcomes.

4.2.2 Best model configurations

The table below reports the tuned configurations for Elastic Net, the additive spline model, and random forest. The linear regression baseline is not tuned.

Table 19: Best tuned hyperparameters for the cancer-type-specific model

	Cancer outcome	Model	Best parameters
0	Breast cancer mortality	Elastic Net (tuned)	{'alpha': 0.01, 'l1_ratio': 0.5}
1	Breast cancer mortality	Additive spline ridge (tuned)	{'alpha': 100.0, 'n_knots': 4}
2	Breast cancer mortality	Random forest (tuned)	{'max_depth': 2, 'max_features': '
3	Mouth/oropharynx cancer mortality	Elastic Net (tuned)	{'alpha': 0.01, 'l1_ratio': 0.2}
4	Mouth/oropharynx cancer mortality	Additive spline ridge (tuned)	{'alpha': 0.1, 'n_knots': 6}
5	Mouth/oropharynx cancer mortality	Random forest (tuned)	{'max_depth': 5, 'max_features': '
6	Oesophagus cancer mortality	Elastic Net (tuned)	{'alpha': 0.01, 'l1_ratio': 0.2}

Table 19: Best tuned hyperparameters for the cancer-type-specific model

	Cancer outcome	Model	Best parameters
7	Oesophagus cancer mortality	Additive spline ridge (tuned)	{'alpha': 100.0, 'n_knots': 4}
8	Oesophagus cancer mortality	Random forest (tuned)	{'max_depth': 2, 'max_features': '
9	Colon and rectum cancer mortality	Elastic Net (tuned)	{'alpha': 0.1, 'l1_ratio': 0.8}
10	Colon and rectum cancer mortality	Additive spline ridge (tuned)	{'alpha': 100.0, 'n_knots': 4}
11	Colon and rectum cancer mortality	Random forest (tuned)	{'max_depth': 4, 'max_features': '
12	Liver cancer mortality	Elastic Net (tuned)	{'alpha': 0.01, 'l1_ratio': 0.2}
13	Liver cancer mortality	Additive spline ridge (tuned)	{'alpha': 10.0, 'n_knots': 4}
14	Liver cancer mortality	Random forest (tuned)	{'max_depth': 3, 'max_features': '

4.2.3 Best model by cancer type

Table 20: Best-performing model for each cancer-type-specific mortality outcome unc cross-validation.

	Cancer outcome	Model	Countries used	Train R^2	CV R^2	C
3	Breast cancer mortality	Random forest (tuned)	180	0.1234	-0.0914	0
15	Colon and rectum cancer mortality	Random forest (tuned)	180	0.6382	0.3801	0
19	Liver cancer mortality	Random forest (tuned)	175	0.3427	0.2067	0
7	Mouth/oropharynx cancer mortality	Random forest (tuned)	180	0.2974	0.1043	0
11	Oesophagus cancer mortality	Random forest (tuned)	174	0.2333	0.0279	0

The best-model summary reinforces that model performance is not uniform across cancer outcomes. Random forests are useful across the five tasks, but the results should not be interpreted as meaning that every cancer type is highly predictable. Colon and rectum cancer is the clearest case where the measured features capture a stronger country-level signal, while liver cancer also shows meaningful but more moderate predictive structure. Breast cancer remains difficult to model in this dataset, even when screening-related and metabolic predictors are included. This likely reflects missing variables such as reproductive history, genetic risk, detailed screening quality, and treatment access, as well as the temporal mismatch between cancer-type outcomes and recent predictors.

4.2.4 Model-agnostic feature importance for selected models

For each cancer type, I interpret the selected best model using permutation importance. This keeps the interpretation consistent across linear, spline, and random forest models. A larger positive value means that randomly permuting that predictor increases RMSE more strongly,

so the fitted model relied more heavily on that feature. These values rank model dependence, not direction of association or causal effect.

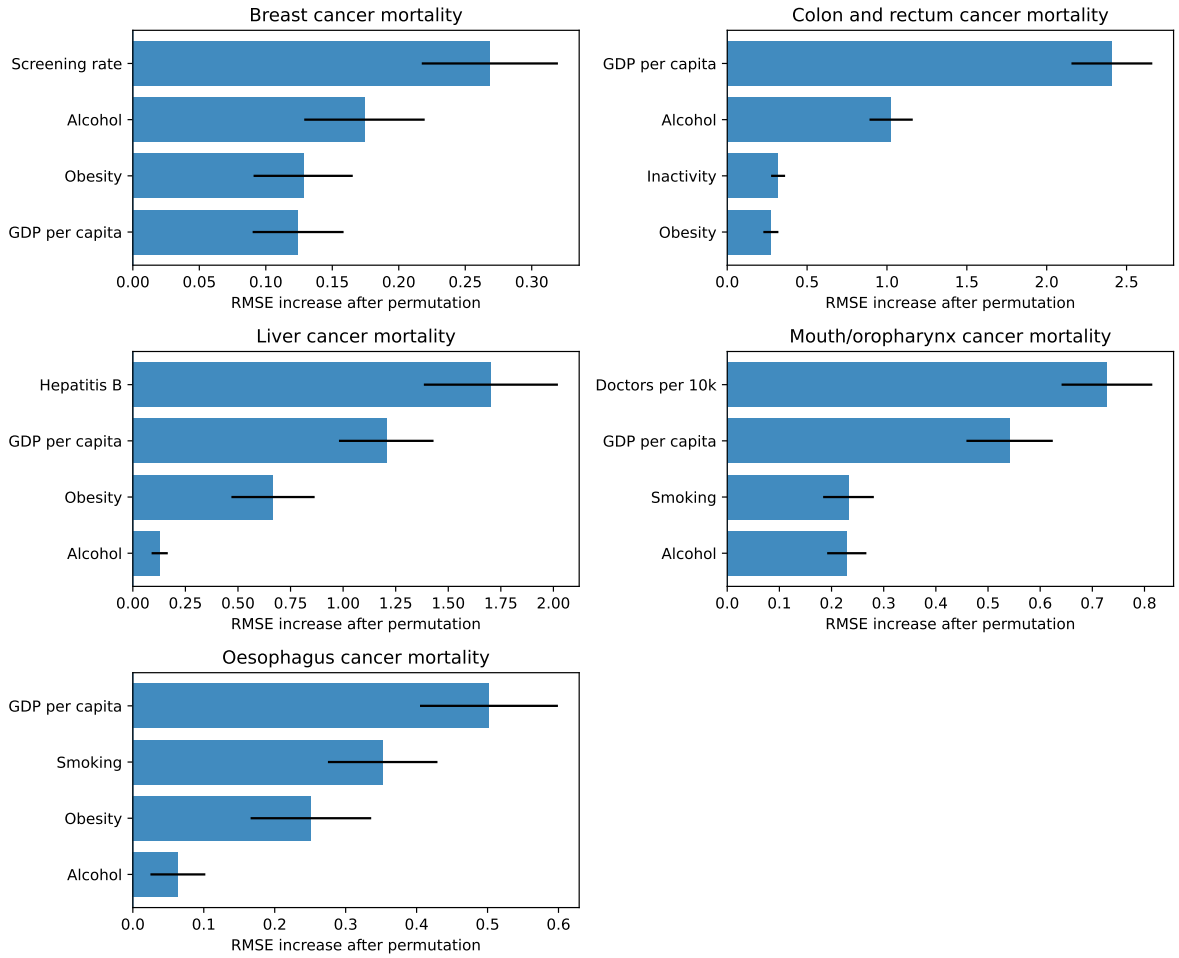


Figure 5: Permutation importance for the selected cancer-type-specific models. Each bar shows the increase in RMSE after the feature is randomly permuted; larger bars indicate that the fitted model relied more strongly on that feature.

4.2.5 Cancer-specific modeling interpretation

This extension changes the modeling question from "Can broad country-level indicators predict overall cancer mortality?" to "Do different cancer-specific mortality outcomes align with different country-level risk profiles?" This is a more realistic framing because cancer mortality is not one homogeneous process. Liver cancer, colorectal cancer, breast cancer, oesophagus cancer, and mouth/oropharynx cancer have different risk-factor structures, so the relevant

predictors should not be identical across outcomes.

The cancer-specific models show that separating cancer types can reveal stronger structure than the aggregate all-cancer model. Colon and rectum cancer and liver cancer show the clearest improvement, suggesting that the selected country-level predictors align better with these outcomes. In contrast, breast cancer and oesophagus cancer remain more difficult to model, which suggests that important drivers are missing or not well represented by the available country-level variables.

The permutation-importance figure provides a descriptive ranking of model reliance. For liver cancer, hepatitis B prevalence is the strongest feature, which is consistent with an infection-related risk mechanism. For colon and rectum cancer, GDP per capita is dominant, with alcohol, physical inactivity, and obesity also contributing; this likely reflects a mixture of development, lifestyle, diet, screening, and reporting differences. For breast cancer, the screening-related variable is ranked highest, but the model's weak cross-validated performance means this should be interpreted cautiously as model reliance on a health-system proxy rather than evidence that screening has a simple directional effect on mortality.

These importance rankings do not provide coefficients. A high importance value means that destroying the information in that feature worsens prediction, not that increasing the feature necessarily increases or decreases cancer mortality. Because many country-level indicators are correlated, variables such as GDP, doctors, and screening may also share explanatory information. The cancer-specific results are therefore best interpreted as exploratory evidence that some mechanisms are more visible for certain cancer types than in the aggregate all-cancer outcome.

5 Conclusions and limitations

This analysis constructed a cross-country dataset linking cancer mortality to lifestyle, environmental, socioeconomic, demographic, healthcare, screening, infection, and urbanization indicators using data from the WHO and World Bank APIs. The results show that broad country-level indicators contain some signal, but that signal is limited and strongly affected by aggregation, missingness, and temporal mismatch.

For the original all-cancer mortality outcome, predictive performance remained modest. The tuned random forest performed best overall, but the improvement over simpler models was small, indicating that model flexibility alone cannot overcome the weak and noisy relationship between the available predictors and aggregate cancer mortality. This supports the interpretation that all-cancer mortality combines many different disease processes into a single outcome, which makes the signal difficult to model using a small set of national indicators.

The cancer-type-specific extension provides a clearer result. Random forests were selected as the best-performing model family across the five cancer-specific tasks, suggesting that nonlinear relationships and interactions are useful in this setting. More importantly, performance varied

substantially by cancer type. Colon and rectum cancer and liver cancer showed stronger cross-validated performance, while breast, oesophagus, and mouth/oropharynx cancer were harder to predict. This supports the idea that separate cancer-type models can recover patterns that are blurred in the aggregate all-cancer outcome.

Feature-importance results also aligned with some expected mechanisms. Hepatitis B prevalence was most important for liver cancer, while colorectal cancer relied strongly on GDP and lifestyle-related variables. Screening-related information was important for breast cancer, although the weak predictive performance for that outcome means the result should be interpreted as model dependence on a health-system proxy rather than a causal screening effect. Across several outcomes, GDP remained highly important, likely because it captures multiple country-level differences at once, including healthcare access, reporting quality, screening systems, diet, and development.

These findings should not be interpreted causally. The cancer-type-specific mortality outcomes available through the WHO API were often older than the 2014–2023 predictor window, so the second modeling stage is exploratory rather than a forward-looking prediction system. In addition, the analysis is ecological: each row represents a country, not an individual patient. Relationships observed at the country level may reflect confounding by age structure, healthcare access, screening, reporting quality, or socioeconomic development.

Several data limitations further constrain the analysis. WHO indicators had uneven country-year coverage, which required using latest available observations and median imputation for predictors. Some important cancer drivers, such as genetic risk, detailed dietary intake, cancer stage at diagnosis, treatment quality, screening follow-up, and historical exposure timing, were unavailable or only crudely proxied. As a result, the models are best viewed as exploratory tools for comparing broad cross-country patterns, not as precise predictive or causal models.

Overall, the project suggests that country-level cancer mortality is difficult to model as a single aggregate outcome. A cancer-type-specific approach is more informative because different cancers are associated with different risk profiles, but even this approach remains limited by data availability and unmeasured confounding. The strongest conclusion is therefore structural: modeling cancer types separately improves interpretability and reveals more meaningful patterns, but high-accuracy prediction would require more temporally aligned, cancer-specific, and clinically detailed data.